2024 Ph H2 Q8

Section: Our Dynamic Universe

Topic: The Expanding Universe (Inverse Square Law & Luminosity)

Summary of Question

Students investigate brightness using an oil spot photometer.

(a) Show that L_2/d_2^2 is constant using the table of values.

(b) Suggest why spherical lamps were used.

(c) Outside, compare lamp and Sun to determine luminosity of the Sun, with mean d, random uncertainty, and calculation.

(a) Establishing the relationship

We calculate L_2/d_2^2 for each result:

- $5.00 / 0.350^2 \approx 40.8$
- $10.00 / 0.495^2 \approx 40.8$
- $15.00 / 0.606^2 \approx 40.9$
- $20.00 / 0.700^2 \approx 40.8$

All values are ≈ 40.8, showing L_2/d_2^2 is constant.

Final statement: $L_2 \propto 1/d_2^2$.

(b) Why small, spherical lamps?

So that each lamp behaves approximately as a point source radiating equally in all directions. This ensures the inverse square law applies accurately.

(c)(i) Mean d

Values: 0.88, 0.86, 0.9, 0.89, 0.86 m

Mean = 0.878 m

Final answer: 0.878 m

(c)(ii) Random uncertainty in mean d

Range = 0.90 – 0.86 = 0.04 m

Half-range = 0.02 m

Uncertainty = ±0.02 m

(c)(iii) Luminosity of the Sun

Formula: $L_{sun} = L_{lamp} \times (D/d)^2$

$= 1.0\times10^{2} \times (1.5\times10^{11} / 0.878)^2$

≈ 2.92e+24 W

Final answer: ≈ 3.0×10^{26} W

Revision Tips

Inverse square law: brightness $\propto 1/d^2$ for a point source.

For experimental data, show consistency by calculating and comparing ratios.

Always quote mean and random uncertainty clearly with units.

Remember typical Sun luminosity is $\sim 3.8 \times 10^{26}$ W; answers of that order are expected.